Banking Loan Risk Analysis Application

REST API Reference Manual & Schema Documentation

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1. Overview & Authentication

The FastAPI service is deployed at /api/v1. Protected endpoints require a Bearer JWT token in the Authorization header. Roles supported: **analyst** and **administrator**.

Authentication Endpoint: POST /api/v1/auth/login (Form data: username, password). Pre-configured accounts: admin/admin123, analyst/analyst123.

2. Customer Management Endpoints

- **GET /api/v1/customers:** Retrieves paginated customer records. Query params: page, page_size, risk_band, education, age_group, gender, search_id, sort_by, order.
- **GET /api/v1/customers/{id}:** Retrieves demographic details and credit profile for a single customer.
- **GET /api/v1/customers/{id}/details:** Retrieves full record including 6-month billing statements and payment history.
- **POST /api/v1/customers:** Creates new customer and initializes credit profile. (Administrator only)
- **PUT /api/v1/customers/{id}:** Updates customer demographic attributes.
- **DELETE /api/v1/customers/{id}:** Removes customer and cascades to bills/payments. (Administrator only)
- **GET /api/v1/customers/export/csv:** Streams customer portfolio in CSV format.

3. Risk Analysis & Underwriting Endpoints

- **POST /api/v1/risk/evaluate:** Evaluates arbitrary financial parameters, returning risk score, band, and triggered business rules.
- **GET /api/v1/risk/review-queue:** Lists all accounts requiring manual underwriter review.
- **PUT /api/v1/risk/review-queue/{customer_id}/status:** Updates underwriter decision (APPROVED, REJECTED, PENDING) with review notes.

4. Analytics & Reporting Endpoints

- **GET /api/v1/analytics/kpis:** Total exposure, average limit, portfolio default rate, utilization, and high-risk percentage.
- **GET /api/v1/analytics/segments:** Segment-level aggregations grouped by education, age group, marital status, and gender.
- **GET /api/v1/analytics/trends:** 6-month historical billing, repayment cashflow, and delinquency rate trends.

5. ETL & Auditing Endpoints

- **POST /api/v1/etl/run:** Triggers ETL pipeline (modes: sample, incremental, full).
- **GET /api/v1/etl/reconciliation:** Returns cryptographic reconciliation audit logs.
- **GET /api/v1/etl/data-quality:** Returns detailed records of automated data quality tests.